

HIMITSU PROTOCOL

Swap from a private note.

Himitsu proves the note spend in your browser, trades through Uniswap V2, and saves the output in a new private note.

PROTOCOL PRESENTATION • 2026



THE PROBLEM

A public wallet swap can link a user to a trade.

A standard wallet swap publishes the sender, amount, and time.

Those public details can connect a wallet to its later activity.

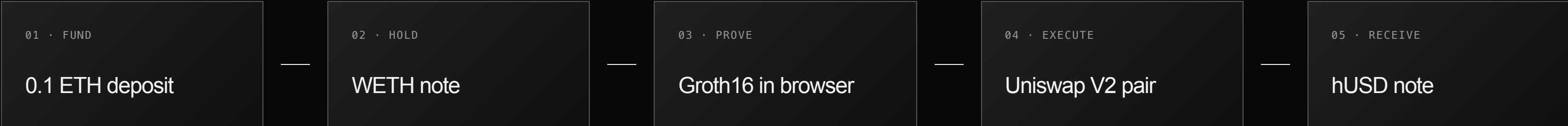
A private note can hide which pool deposit is spent, but it needs proof-based authorization to trade through an AMM.

The product problem is to authorize the note, execute a limited swap, and place the output in a new note.

THE SOLUTION

Himitsu swaps from a private note through Uniswap V2.

The browser proves ownership. The pool checks the allowed action. The AMM executes the swap. A second pool records the output as a new private note.



The user saves the output note and uses it to withdraw or make another swap.

HOW THE SWAP WORKS

The user controls both private notes.

INPUT NOTE · WETH

0.1 WETH

秘密 · held by you

The browser creates the output note before it submits the transaction. After confirmation, the app checks public pool events to recover the output amount.

OUTPUT NOTE · hUSD

actual AMM output

秘密 · recovered by you

The browser creates the proof and submits the transaction to the configured node. The app does not send the note to a hosted proving service or relayer.

WHY THE CURRENT FLOW DOES NOT USE ERC-4337

Himitsu submits a native EIP-8141 transaction.



EIP-8141 TRANSACTION TYPE

ERC-4337 sends a UserOperation through an EntryPoint contract and a bundler. Himitsu does not create UserOperations or use that route.

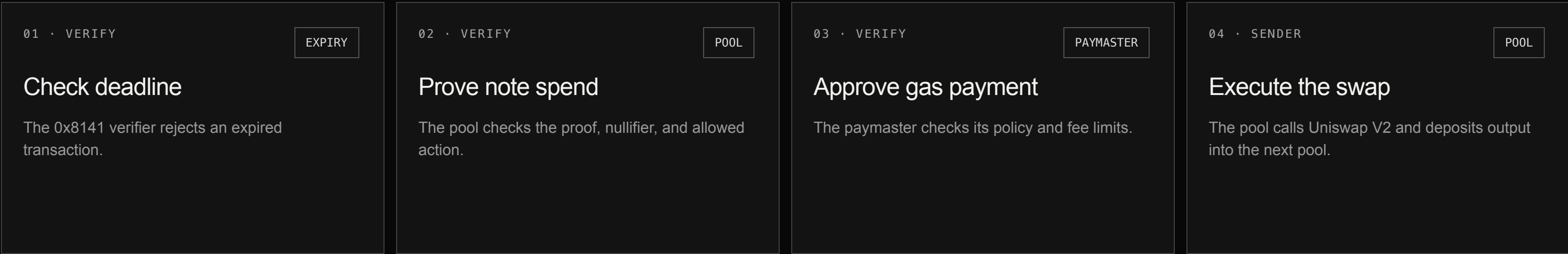
The current design needs the shared pool to be the transaction sender, prove the note spend, and authorize a separate gas payer. EIP-8141 represents those steps as ordered frames in one transaction.

This is an architecture choice for an EIP-8141-compatible Ethrex network. ERC-4337 can also support sponsored accounts, but it uses a different transaction path.

Sources: [EIP-4337](#) · [EIP-8141 draft](#)

EIP-8141 IS THE TRANSACTION CORE

One type-0x06 transaction runs four ordered frames.



The pool is the transaction sender. The separately funded paymaster pays network gas. The user’s wallet does not pay gas for the swap.

WHY THIS FITS THE PRODUCT

Frame transactions put sender, validation, and payment in one flow.

APPROACH	TRANSACTION PATH	TRADE-OFF FOR HIMITSU
Direct wallet swap	The user’s EOA sends the swap and pays gas.	The wallet address is the sender. Himitsu instead uses the pool as sender and a prefunded paymaster for gas.
Relayer meta-transaction	An app service receives a signed request and submits a transaction.	The relayer adds a service and signing-key dependency. Himitsu’s browser submits the raw frame transaction to the node.
ERC-4337	A bundler submits a UserOperation through an EntryPoint.	It supports sponsored smart accounts, but adds the UserOperation and EntryPoint path that this pool-sender design does not need.
EIP-8141 in Himitsu	One type-0x06 transaction orders pool validation, gas approval, and execution.	It fits the shared pool sender and separates proof authorization from paymaster approval. It requires network support for EIP-8141.

Sources: [EIP-4337](#) · [EIP-8141 draft](#)

OTHER PRIVACY SYSTEMS SOLVE RELATED PROBLEMS

Himitsu focuses on one private swap route.

PROTOCOL	WHAT IT DOES	HIMITSU'S FOCUS
Tornado Cash Classic	Users deposit fixed amounts and later withdraw with a zero-knowledge proof.	Himitsu also spends a note through an AMM and records the output in a second pool.
RAILGUN	Users hold shielded balances and can make private calls to other contracts.	Himitsu focuses on a shared WETH and hUSD pool, with EIP-8141 frames for note validation and gas payment.
Himitsu	A browser-generated proof authorizes a note spend that calls Uniswap V2.	The output is committed to a new private note in the same transaction.

Himitsu is not a replacement for broader privacy systems. Its contribution is this transaction design and its EIP-8141 implementation.

Sources: [Tornado Cash Classic](#) · [RAILGUN cross-contract calls](#) · [EIP-8141 draft](#)

WHAT THE PROOF HIDES, AND WHAT IT DOES NOT

The proof hides the spent note, not the public swap.

Private

The proof shows that the user controls a valid, unspent note without naming its deposit commitment. The note secret and witness stay in the browser.

NOTE SECRET

SPEND MEMBERSHIP

BROWSER PROOF

Public

Deposit and swap amounts, transaction timing, AMM activity, recipient addresses, and RPC metadata remain observable. These details can still support correlation.

AMOUNTS

TIMING

RPC DATA

Himitsu does not hide amounts, timing, or network metadata.

WHAT EXISTS TODAY

The application implements the full note-to-note flow.

Implemented

The browser creates notes and Groth16 proofs. The pool validates spends. A paymaster applies gas limits. The frame transaction swaps through Uniswap V2 and creates an output note.

The repository includes integration flows for deposits, swaps, withdrawals, and failed swaps.

Not yet established

EIP-8141 is still a draft. The current deployment uses a pinned Ethrex implementation. The contracts and circuits need independent security review.

Public-network compatibility, throughput, sponsor operations, and privacy under real user activity remain to be evaluated.

NEXT VALIDATION STEPS

We need to validate the network, security, and user experience.

Run the full transaction on a publicly reachable EIP-8141 network.
Complete independent reviews of the contracts, circuit, and setup.
Measure how users create, store, and recover notes.